Shih-Ni Prim

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EDUCATION

PhD North Carolina State University, Statistics 2022-present
Advisor: Brian J. Reich
Cumulative GPA: 4.0
Expected graduation date: May 2026

MR North Carolina State University, Statistics Dec 2022
Cumulative GPA: 4.0

COMPUTER SKILLS

Programming: R, Python
Certificate: SAS Certified Specialist: Base Programming Using SAS 9.4 (Issued May 2020)
Tools: High performance computing, Git version control, parallel computing

AWARDS

B. B. Bhattacharyya Excellence Award 2023–2024
North Carolina State University, Department of Statistics

RESEARCH INTERESTS AND EXPERIENCES

Bayesian inference and hierarchical modeling, spatial and spatiotemporal statistics, tensor decomposition, tensor regression, surrogate modeling for computer experiment, Gaussian process, active learning, climate and environmental applications, causal inference

RESEARCH EXPERIENCE

Environmental Protection Agency, Chapel Hill, NC August 2024 to present
Oak Ridge Institute for Science and Education (ORISE) Fellow

- Advisors: Ana Rappold and Brian Reich
- Research project: A Multivariate Spectral Model with Adjustment for Unmeasured Confounders for Multiple Outcomes and Multiple Exposures
- Proposed a spectral, fully Bayesian model to adjust for spatial confounding and ensure causal interpretation of effects
- Constructed fully Bayesian MCMC algorithms in R with different priors and capacity (multivariate vs univariate outcomes) to compare model performance
- Used tensor structure so that coefficients can vary by spatial scales, exposures, and outcomes
- Implemented CP decomposition to reduce dimension of the tensor
- Used shrinkage and half Cauchy prior to encourage sparsity
- Performed extensive simulation studies
- (The tasks above started in January 2023.)
- Applied proposed model on Per- and polyfluoroalkyl substances (PFAS) data
- Visualized Geographic Information System data using R (sf, tidyverse, and ggplot2)
- Currently drafting manuscript for publication

Collaborator of Lawrence Livermore National Laboratory August 2024 to present

- Advisors: Kevin Quinlan and Annie Booth
- Research project: Active Learning with Deep Gaussian Processes for Aero Database Construction
- Examined predictive accuracy of Deep Gaussian Processes versus ordinary Gaussian Processes as surrogate models for aerodynamic data
- Devised active learning algorithms to efficiently construct aero databases in Python (scikit-learn, numpy, dgpsi)
- Created visualization of the sequential sampling process and results in Python (matplotlib)
- Presented preliminary results at conference

Lawrence Livermore National Laboratory, Livermore, CA May to August 2024
Data Science Summer Institute Intern

- Conducted literature review on surrogate modeling, contour estimation, and active learning
- Presented at end-of-summer presentation for interns

North Carolina State University, Raleigh, NC August 2022 to May 2024
Research Trainee at Duke Clinical Research Institute

- Helped with developing and evaluating analytic methods based on WIN statistics
- Shadowed meetings of clinical research and received training on ethics of research

North Carolina State University, Raleigh, NC Summer 2022, July 2023 to May 2024
Research Assistant

- Principal Investigators: Natalie Nelson and Brian Reich
- Used statistical and machine learning models to analyze spatiotemporal data to examine inference about harmful algal blooms
- Data cleaning and preprocessing, data analysis, statistical modeling, manuscript drafting

Wake Forest Baptist Health, Winston-Salem, NC 2021 to 2022
Volunteer Research Statistician / Intern

- Conducted Kaplan Meier Survival Analysis and Cox Proportional Hazard Model on retrospective data of patients with glioblastoma
- Created tables and plots and help with writing and editing for publication
- Helped collect MRI images and clinical data for studies that utilize machine learning for classification

PUBLICATIONS

Chan MD, **Prim S**, Cramer C, Ruiz J. “Nonrandomized trials in clinical oncology.” In *Handbook for Designing and Conducting Clinical and Translational Research: Translational Radiation Oncology*, edited by Adam E. M. Eltorai, Jeffrey A. Bakal, Daniel W. Kim, David E. Wazer, 313–284. 2023. Academic Press.

Helis CA, **Prim S**, Cramer CK, Strowd R, Lesser GJ, White JJ, Tatter SB, Laxton AW, Whitlow C, Lo H, Debinski W, Ververs JD, Black PJ, Chan MD, “Clinical outcomes of dose-escalated re-irradiation in patients with recurrent high-grade glioma.” *Neuro-Oncology Practice* (2022). https://doi.org/10.1093/nop/npac032.

Prim S. “The Utah Symphony’s Recordings with Vanguard Records.” *Association for Recorded Sound Collections Journal* 48/1 (2017): 23–41.

CONFERENCE PRESENTATIONS

Prim S., Quinlan K, Booth A. "Active Learning with Deep Gaussian Processes for Aero Database Construction." International Conference on Advances in Interdisciplinary Statistics and Combinatorics, October 13, 2024.

Prim S. "The Utah Symphony's 1971 Goodwill Tour in Latin America: Cultural Diplomacy during the Cold War." American Musicological Society Southeast Chapter Spring Meeting, Furman University, Greenville, SC, March 18, 2017.

Prim S. "Maurice Abravanel and Gustav Mahler: The Reception of Early Mahler Recordings by Abravanel and the Utah Symphony Orchestra." American Musicological Society Southeast Chapter Spring Meeting, University of North Carolina at Chapel Hill, NC, February 23, 2014.

PROFESSIONAL SERVICE

Book Reviewer — *Notes* (2018 to 2021)
Reviewer — *Writing Center Journal* (2015 to 2016)
Grant Reviewer — University of Iowa (2013)

GRANTS AND FELLOWSHIPS

ORISE Fellowship — 2024 to present
Environmental Protection Agency

T32 Training Grant — 2022 to 2024
Duke Clinical Research Institute

LANGUAGES

English, Mandarin, German, French, Italian

REFERENCES

Brian Reich, Gertrude M Cox Distinguished Professor of Statistics
Department of Statistics, North Carolina State University
Phone: 919-513-7686
Email: bjreich @ncsu.edu

Yawen Guan, Assistant Professor
Department of Statistics, Colorado State University
Email: yawen.guan@colostate.edu

Natalie Nelson, Associate Professor
Department of Biological and Agricultural Engineering, North Carolina State University
Phone: 919-515-6741
Email: nnelson4@ncsu.edu

OTHER EDUCATION

PhD University of Iowa, Musicology May 2016
Dissertation: "Maurice Abravanel and the Utah Symphony's Performances and Recordings of Gustav Mahler's Symphonies (1951–1979)" (Advisor: Nathan Platte)

MM Florida State University, Musicology May 2009
Thesis: "Gustav Mahler's *Das Lied von der Erde*: An Intellectual Journey across Cultures and Beyond Life and Death" (Advisor: Douglass Seaton)

MM University of North Carolina at Greensboro, Saxophone Performance May 2006

MA University of North Carolina at Chapel Hill, Comparative Literature May 2004
Thesis: "Music-Myth into Comedy: How Shaw Used Wagner to Transform Mozart's Legend" (Advisor: William Harmon)

BA National Chiao Tung University, Foreign Languages and Literatures June 2000

TEACHING EXPERIENCE

North Carolina State University, Raleigh, NC May to August 2022
Teaching Assistant, Department of Statistics
- Assist with two graduate courses: Fundamentals of Statistical Inference I and Fundamentals of Linear Models and Regression
- Duties: Grade homework and hold weekly office hours

Wake Forest University, Winston-Salem, NC August 2017 to December 2017
Adjunct Faculty, Department of Music
- Independently designed and taught the course "Introduction to Western Music," which covered western music history from the Middle Ages to the Twentieth Century
- Used music as artifacts to help students think about history in the appropriate contexts and cultivate critical thinking
- Provided necessary study aids including detailed guidelines and rubrics for writing assignments and study guides for exams
- Used a scaffolding method to help students write research papers

University of Iowa, Iowa City, IA 2011 to 2016
Writing Tutor, Writing Center
- Online tutoring (Fall 2013–Spring 2016)
    - Provided feedback for papers through online tutoring (10 hours per week)
- Online and Face-To-Face tutoring (Fall 2011, Spring 2012)
    - Met with 4 students for 4 hours of in-person tutoring sessions per week
    - Provided feedback for papers through online tutoring